Kakarla Venkata Rama Sai Pavan Shanmukh

+91 6302509027 | pavanshanmukhkakarla@gmail.com | GitHub | LinkedIn

Career Objective

Adaptable B.Tech student and full-stack developer specializing in scalable distributed ledger analytics and secure real-time systems to drive data-driven architectural innovation.

Education

KL University | 9.51 CGPA

Aug 2023 - Present

Bachelor of Technology (B.Tech) | Computer Science and Information Technology — Vijayawada A.P

Narayana Educational Institutions | 9.31 CGPA

Aug 2021 - Jul 2023

Intermediate | Andhra Pradesh Board of Intermediate Education (BIEAP) — Vijayawada A.P

Nalanda Vidya Niketan | 8.18 CGPA

Jul 2020 - May 2021

10th Grade | Central Board of Secondary Education (CBSE) — Vijayawada A.P

Skills

Core: Data Structures and Algorithms, DBMS, Operating Systems, Computer Networks, Linux, Git and GitHub, OOPS in Java

Languages: Java, Python, HTML5, CSS3, JavaScript

Tools, Frameworks and Databases: React.js, Spring Boot, Node.js, Power BI, CI/CD pipelines, MySQL, PostgreSQL, Postman

Experience

MeshaSec Technologies Pvt Ltd.

May 2026 - Jul 2026

Cybersecurity Research & AI Intern

Gained hands-on experience in web application security and backend development by performing vulnerability assessments using DAST tools, solving 10+ security lab machines, and developing/testing REST APIs using Node.js, Express.js, and Postman.

Cisco Networking Academy

May 2024 - Jul 2024

Intern

Hands-on networking with Cisco Packet Tracer. Designed and simulated network topologies; implemented and tested routing algorithms for optimized performance.

Projects

SelectShans: Ransomware Pre-Encryption Detection System

Feb 2026 - Oct 2026

Abstract: Real-time system detecting ransomware through suspicious file activity.
Technologies: Python, Flask, React.js, MySQL, JWT, REST APIs, Docker.
Outcome: Provides alerts, secure logs, forensic reports, and data-loss protection.
Technologies: Python, React.js, Network Security, JavaScript, Git, GitHub, Docker

OmniBank: Full-Stack Financial Management System

Oct 2025 - Mar 2026

Abstract: Secure full-stack banking platform with real-time transactions and MFA.
Technologies: React.js, Vite, Spring Boot, Spring Security, MySQL, REST APIs, CI/CD.
Outcome: Enables secure and transparent financial tracking.
Technologies: JavaScript, CI / CD, Docker, Kubernetes, Jenkins, React.js, AWS Cloud

LexiTree: Optimized Text Analysis via Self-Balancing AVL Trees

Nov 2023 - Feb 2024

Abstract: Optimized AVL Tree-based word frequency analysis to O(log n) for large datasets.
Technologies: AVL Tree, UI Frameworks, Performance Analysis.
Outcome: Improved large-scale data processing.
Technologies: React.js, Data Structures and Algorithms, Git, GitHub, Web Technologies

Certifications and Achievements

Microsoft Certified Azure Fundamentals Mar 2026

Acquired basic knowledge of cloud services and how they are provided via Microsoft Azure.

AWS Certified Developer Associate Jul 2026

Learnt about AWS Lambda, Amazon API Gateway, AWS DynamoDB, AWS S3, SQS, SNS, ECS, CloudWatch, IAM.

Optimized Antenna Design for WBAN Applications Dec 2024 - Present

A high-efficiency, flexible design featuring a metamaterial-backed ground plane to ensure stable on-body communication.

Cambridge University - Linguaskill May 2024 - Present

Developed strong English speaking, listening, vocabulary, pronunciation, and communication skills, enabling me to express ideas clearly and confidently in professional and everyday situations.

Microsoft Certified: GitHub Foundations (GH-900) Oct 2026

Demonstrated foundational knowledge of GitHub, including Git and version control, repositories, collaboration, GitHub Actions, and GitHub Copilot.

Coding Profiles

Leetcode: PavanShanmukhKakarla Codechef: kakarla_pavan1 HackerRank: h2300090201 CodeForces: 2300090201